

(12) $\triangle ABC \sim \triangle DEC$

$$\frac{BC}{EC} = \frac{AB}{DE}$$

$$\Rightarrow \frac{10}{7} = \frac{AB}{5}$$

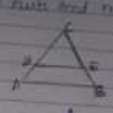
$$\Rightarrow AB = \frac{50}{7}$$

$$AB = 7.14 \text{ miles}$$

(13) Given, $AC = 5 + BC$
 $BC = 10 - 5$
 $= 5 \text{ miles}$

Now, $CF = \frac{7}{10} \times AC$
 $= \frac{7}{10} \times 5$
 $= 3.5 \text{ miles}$

(14) $\triangle ABC$ is a triangle formed by both hills and mountain top.
 $\triangle DEC$ is the triangle formed by both hills and mountain top.



$DE \parallel AB$
 SAT

In $\triangle ABC$ and $\triangle DEC$
 $\angle ACB = \angle DEC$ (Common)
 $\angle CAB = \angle CDE$ (Corresponding angles formed by $DE \parallel AB$)

$\therefore \triangle ABC \sim \triangle DEC$ (By AA) And Hence

$$\frac{BC}{EC} = \frac{AB}{DE}$$

$$\frac{10}{7} = \frac{AB}{5}$$

(15) Since $DE \parallel AB$
 $\frac{CD}{AD} = \frac{CE}{EB}$ (Corresponding sides)

$$\Rightarrow \frac{10}{20} = \frac{7}{EB}$$

$$\Rightarrow EB = 14$$

$BD = 14 + 10 = 24$

$$\frac{PA}{CA} = \frac{PB}{CB}$$

$$\Rightarrow \frac{2}{y} = \frac{1}{x+y}$$

(1) $\frac{2}{x} + \frac{2}{y} = \frac{1}{x+y} + \frac{1}{x+y}$

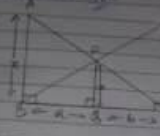
$$\Rightarrow \frac{2}{x} + \frac{2}{y} = \frac{2}{x+y}$$

$$\Rightarrow \frac{1}{x} + \frac{1}{y} = \frac{1}{x+y}$$

[Multiplying both sides by $\frac{1}{2}$]

Hence proved

Answer: 20 - 8h



Sol: In fig $\angle AED = \angle BEC$ (Vertically opposite angles)

$AB = 20 \text{ units}$
 $AD = 8 \text{ units}$
 $BC = 8 \text{ units}$

To find $= \frac{1}{x} + \frac{1}{y} = \frac{1}{z}$

Proof: In $\triangle AED$
 Let $AE = a$ units
 $ED = b$ units

In $\triangle AED$ and $\triangle BEC$
 $\angle AED = \angle BEC$ (Each 90°)
 $\angle EAD = \angle EBC$ (Common)
 $\therefore \triangle AED \sim \triangle BEC$ (By AA)

$$\frac{AE}{BE} = \frac{ED}{EC}$$

$$\Rightarrow \frac{a}{b} = \frac{b}{a+b}$$

Ques 24

$$2x + 2y = 70 \Rightarrow x + y = 35$$

$$2x + y = 45$$

$$\begin{array}{r} 2x + 2y = 70 \\ - (2x + y = 45) \\ \hline y = 25 \end{array}$$

Putting $y = 25$ in (i)

$$x + 25 = 35$$

$$\Rightarrow x = 35 - 25 = 10$$

$$\Rightarrow x = 10$$

Father's age = 30 years
Son's age = 15 years

Answer 25: Sol

Let Ratio be $k:1$
A(1, -7), B(6, 1)

$$k = \frac{6-1}{1-(-7)} = \frac{5}{8}$$

$$\Rightarrow \frac{1-k}{1+k} = 0$$

$$\Rightarrow \frac{1-k}{1+k} = 0$$

$$\Rightarrow 1-k = 0$$

$$\Rightarrow k = 1$$

Ratio = 1:1

Answer 24: Sol

Let father's age be x years
Son's age be y years

Acc. to 1st condition
 $2x + 2y = 70 \rightarrow (i)$

Acc. to 2nd condition
 $2x + y = 45 \rightarrow (ii)$

(i) $\times 2 \Rightarrow 2x + 4y = 140 \rightarrow (iii)$

(ii) $\times 1 \Rightarrow 2x + y = 45 \rightarrow (iv)$

C will complete one round in = 200/1000
= 0.2 min

Let A, B, & C start at time $t=0$ and they would have collectively completed n_1, n_2 and n_3 rounds and they would be at a distance from starting point

$$200t = 1000n_1, x = 1000n_1 \rightarrow (i)$$

$$100t = x + 1000n_2 \rightarrow (ii)$$

$$200t = x + 1000n_3 \rightarrow (iii)$$

Substituting x from eq (i) in (ii) and (iii)

$$100t = 200t - 1000n_1 + 1000n_2$$

$$1000(n_2 - n_1) = 100t$$

$$t = 10(n_2 - n_1)$$

Also,

$$200t = 200t - 1000n_1 + 1000n_3$$

$$1000(n_3 - n_1) = 100t$$

$$10(n_3 - n_1) = t$$

t is same as t and n_1, n_2, n_3 can be whole number.

LCM of 10 and 10 = 10

they meet at 10 min.

Midpoint of AB = $\left(\frac{2+6}{2}, \frac{-7+1}{2}\right)$

$$\frac{2+x}{2} = \frac{7}{2}, \frac{-7+y}{2} = \frac{1}{2}$$

$$\Rightarrow \frac{2+x}{2} = \frac{7}{2}$$

$$\Rightarrow 2+x = 7$$

$$\Rightarrow x = 5$$

$$\frac{-7+y}{2} = \frac{1}{2}$$

$$\Rightarrow -7+y = 1$$

$$\Rightarrow y = 8$$

The coordinates of fourth vertex = D(5, 8)

Answer 23: Sol

Unemployment of path = 1000m

A will complete one round in = 1000/1000 = 1 min

B will " " " = 1000/1000 = 1 min

C will " " " = 1000/1000 = 1 min

Distance = 100 km
 Let the speed of car A be x km/h
 B be y km/h

Same direction
 Speed = $x - y$
 Distance = 100 km
 $t = 5$ hours
 $100 = (x - y) \cdot 5$
 $x - y = 20 \quad \text{--- (1)}$

Opposite direction
 Speed = $x + y$
 Distance = 100 km
 $t = 1$ hr
 $100 = (x + y) \cdot 1$
 $x + y = 100 \quad \text{--- (2)}$

Solving (1) and (2)
 $x - y = 20$
 $x + y = 100$
 $\hline 2x = 120$
 $x = 60$ km/h

Let A & B be the points on the line
 whose coordinates are $(1, 0)$ and $(3, 0)$
 The line is AB
 $(x - 1)^2 + (y - 0)^2 = (x - 3)^2 + (y - 0)^2$
 $\Rightarrow (x - 1)^2 - (x - 3)^2 = 0$
 $\Rightarrow x^2 - 2x + 1 - (x^2 - 6x + 9) = 0$
 $\Rightarrow x^2 - 2x + 1 - x^2 + 6x - 9 = 0$
 $\Rightarrow 4x - 8 = 0$
 $\Rightarrow x = 2$
 Hence the point is $(2, 0)$

Mid point of AB
 $PK = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$
 $= \left(\frac{1 + 3}{2}, \frac{0 + 0}{2} \right)$
 $= \left(\frac{4}{2}, \frac{0}{2} \right)$
 $= (2, 0)$

Distance = 100 km
 Let the speed of car A be x km/h
 B be y km/h

Same direction
 Speed = $x - y$
 Distance = 100 km
 $t = 5$ hours
 $100 = (x - y) \cdot 5$
 $x - y = 20 \quad \text{--- (1)}$

Opposite direction
 Speed = $x + y$
 Distance = 100 km
 $t = 1$ hr
 $100 = (x + y) \cdot 1$
 $x + y = 100 \quad \text{--- (2)}$

Solving (1) and (2)
 $x - y = 20$
 $x + y = 100$
 $\hline 2x = 120$
 $x = 60$ km/h

Distance = 100 km
 Let the speed of car A be x km/h
 B be y km/h

Same direction
 Speed = $x - y$
 Distance = 100 km
 $t = 5$ hours
 $100 = (x - y) \cdot 5$
 $x - y = 20 \quad \text{--- (1)}$

Opposite direction
 Speed = $x + y$
 Distance = 100 km
 $t = 1$ hr
 $100 = (x + y) \cdot 1$
 $x + y = 100 \quad \text{--- (2)}$

Solving (1) and (2)
 $x - y = 20$
 $x + y = 100$
 $\hline 2x = 120$
 $x = 60$ km/h

MATHEMATICS

10-09-2023

Ansno 1: (1) 3

Ansno 2: (4) No solution

Ansno 3:- (1) 10 units

Ansno 4:- (2) Proportional

2

Ans no 5:- (2) $-\frac{1}{2}$

Ans no 6: (1) x^3y^3

Ans no 7: Soln

$$4x^2 - 2x + (K-4)$$

$$\alpha + \beta = -\left(\frac{-2}{4}\right)$$

$$= \frac{2}{4}$$

$$= \frac{1}{2} \rightarrow \textcircled{I}$$

$$\alpha \cdot \beta = \frac{K-4}{4} \rightarrow \textcircled{II}$$

$$\alpha + \beta = \frac{1}{2} (\alpha \cdot \beta)$$

$$\Rightarrow \frac{1}{2} = \frac{1}{2} \times \left(\frac{K-4}{4}\right) \quad \left[\text{From } \textcircled{I} \& \textcircled{II} \right]$$

$$\Rightarrow \cancel{\frac{1}{2}} \cdot \frac{1}{\cancel{2}} \times \left(\frac{K-4}{4}\right) = \frac{1}{2}$$

$$\Rightarrow \frac{K-4}{8} = \frac{1}{2}$$

$$\Rightarrow 2K - 8 = 8$$

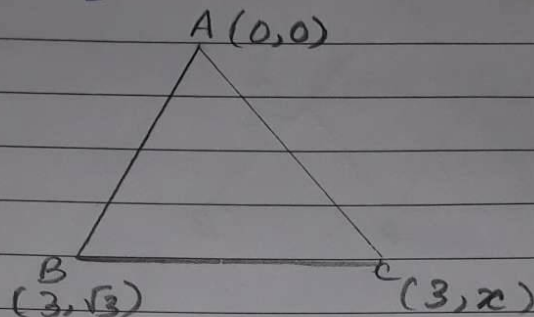
P.T.O

$$\Rightarrow 2K - 8 = 8$$

$$\Rightarrow 2K = 16$$

$$\therefore K = 8$$

Ans no 8:- Sol



$A(0,0)$, $B(3,\sqrt{3})$, $C(3,x)$ forms an equilateral triangle.

$\therefore$ The triangle is equilateral. So,

$$(AB)^2 = (AC)^2$$

$$\Rightarrow (3-0)^2 + (\sqrt{3}-0)^2 = (3-0)^2 + (x-0)^2$$

$$\Rightarrow (\sqrt{3})^2 = x^2$$

$$\Rightarrow 3 = x^2$$

$$\therefore x = \sqrt{3}$$

Ans no 9:-

$$2px + 3y = 7$$

$$2x + y = 6$$

$$\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$$

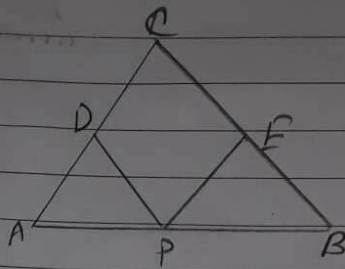
$$\Rightarrow \frac{2p}{2} \neq \frac{3}{1}$$

$$\therefore p \neq 3$$

Any real value for except 3, the give pair of

Linear equations will have exactly one solution.

Ansno 10:



Solⁿ: In the figure it is given that $PD \parallel BC$ and $\frac{AD}{DC} = \frac{CE}{BE}$

To Prove :- $PE \parallel AC$

Proof:

$$\begin{array}{l} PD \parallel BC \quad [\text{given}] \\ \frac{AD}{DC} = \frac{CE}{BE} \end{array}$$

According to BPT $\rightarrow$ (i)

$\therefore PD \parallel BC$

$$\therefore \frac{AD}{DC} = \frac{AP}{PB} \rightarrow (ii)$$

From (i) and (ii)

$$\frac{AP}{PB} = \frac{CE}{BE}$$

$\therefore$

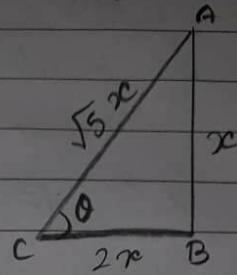
$\therefore PE \parallel AC$ [Acc to converse BPT]

PTD

Ansoll:-

Given,

$$\tan \theta = \frac{1}{2}$$



To Prove $= \frac{1}{2} \sin \theta \cos \theta = \frac{1}{5}$

Proof -

$$\tan \theta = \frac{1}{2}$$

$$\Rightarrow \tan \theta =$$

$$\Rightarrow \frac{AB}{BC} = \frac{1}{2}$$

Let $AB = 'x'$
 $BC = '2x'$

$$\sin \theta =$$

Now,

$$(AB)^2 + (BC)^2 = (AC)^2$$

$$\Rightarrow (AC)^2 = x^2 + (2x)^2$$

$$\Rightarrow AC = \sqrt{x^2 + 4x^2}$$

$$\Rightarrow AC = \sqrt{5x^2}$$

$$\Rightarrow AC = \sqrt{5} x$$

Now

$$\sin \theta = \frac{x}{\sqrt{5} x} = \frac{1}{\sqrt{5}}$$

$$\cos \theta = \frac{2x}{\sqrt{5} x} = \frac{2}{\sqrt{5}}$$

PTO

Now, LHS

$$\frac{1}{2} \sin \theta = \cos \theta$$

$$= \frac{1}{2} \times \frac{1}{\sqrt{5}} \times \frac{2}{\sqrt{5}}$$

$$= \frac{1}{5}$$

$\therefore$ LHS = RHS
Hence, Proved

Ansno 12:- 80h

The fraction = $\frac{257}{5000}$

$$= \frac{257}{5000}$$

Denominator in the form $2^n \times 5^m$

$$5000 = 2 \times 2 \times 2 \times 5 \times 5 \times 5 \times 5$$

$$= 2^3 \times 5^4$$

$$\frac{257}{2^3 \times 5^4}$$

$$= \frac{257 \times 2}{2^3 \times 5^4 \times 2}$$

$$= \frac{514}{10000}$$

$$= 0.0514$$

[Multiplying numerator & denominator by 2]

$$\begin{array}{r} 2 \overline{) 5000} \\ \underline{2000} \\ 3000 \\ \underline{2000} \\ 1000 \\ \underline{2000} \\ 0 \end{array}$$

$$\begin{array}{r} 2 \overline{) 2500} \\ \underline{2000} \\ 500 \\ \underline{2000} \\ 0 \end{array}$$

$$\begin{array}{r} 2 \overline{) 250} \\ \underline{200} \\ 50 \\ \underline{200} \\ 0 \end{array}$$

$$\begin{array}{r} 5 \overline{) 125} \\ \underline{100} \\ 25 \\ \underline{25} \\ 0 \end{array}$$

$$\begin{array}{r} 5 \overline{) 25} \\ \underline{25} \\ 0 \end{array}$$

$$\begin{array}{r} 257 \\ 2 \overline{) 514} \\ \underline{514} \\ 0 \end{array}$$

$$\begin{array}{r} (10) \overline{) 10000} \\ \underline{10000} \\ 0 \end{array}$$

Ansno 13:-

Let us assume $\sqrt{3}$ as a rational number

$\therefore \sqrt{3} = \frac{a}{b}$, a and b are co-primes having 1 as a common factor and $q \neq 0$

$$\Rightarrow (\sqrt{3})^2 = \frac{a^2}{b^2} \quad [\text{squaring both sides}]$$

$$\Rightarrow 3 = \frac{a^2}{b^2}$$

$$\Rightarrow a^2 = 3b^2 \rightarrow \textcircled{I}$$

Hence 3 divides a^2

$\therefore$ 3 divides $a \rightarrow \textcircled{II}$

$$\frac{a}{3} = c \quad [c \text{ is some integer}]$$

$$\Rightarrow a = 3c$$

$$\Rightarrow a^2 = (3c)^2 \quad [\text{squaring both sides}]$$

$$\Rightarrow 3b^2 = 3c^2 \quad [\text{from } \textcircled{I}]$$

$$\Rightarrow b^2 = 3c^2$$

$$\Rightarrow \frac{b^2}{3} = c^2$$

3 divides b^2

$\therefore$ 3 divides $b \rightarrow \textcircled{III}$

But, 3 divides a & b

From $\textcircled{II}$ & $\textcircled{III}$

3 divides a & b both

$\therefore$ 3 is a factor of a & b

$\therefore a$ & b are not co-prime

Hence, our assumption is wrong

$\therefore$ By contradiction $\sqrt{3}$ is irrational

Ansno 14:-

$$P(x) = 2x^2 - 3x + 7$$

α & β are the zeroes of $P(x)$

$$\therefore \alpha + \beta = -\left(\frac{-3}{2}\right)$$

$$= \frac{3}{2}$$

$$\alpha \cdot \beta = \frac{7}{2}$$

$$\frac{1}{2\alpha - 3} + \frac{1}{2\beta - 3}$$

$$= \frac{(2\beta - 3) + (2\alpha - 3)}{(2\alpha - 3)(2\beta - 3)}$$

$$= \frac{(2\beta - 3) + (2\alpha - 3)}{4\alpha\beta - 6\alpha - 6\beta + 9}$$

$$= \frac{2\beta - 3 + 2\alpha - 3}{4\alpha\beta - 6(\alpha + \beta) + 9}$$

$$= \frac{2(\alpha + \beta) - 6}{4\alpha\beta - 6(\alpha + \beta)}$$

$$= \frac{2(\alpha + \beta) - 6}{4\alpha\beta - 6(\alpha + \beta)}$$

$$= \frac{2\left(\frac{3}{2}\right) - 6}{4\left(\frac{7}{2}\right) - 6\left(\frac{3}{2}\right)}$$

$$= \frac{3 - 6}{14 - 9}$$

$$= \frac{-3}{5}$$

Ans no 15:- Solⁿ

$$3x - (a+1)y = 2b-1$$

$$5x + (1-2a)y = 3b$$

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$$

$$\frac{a_1}{a_2} = \frac{3}{5}$$

$$\frac{b_1}{b_2} = \frac{-(a+1)}{1-2a} = \frac{-a-1}{1-2a}$$

$$\frac{c_1}{c_2} = \frac{2b-1}{3b}$$

$$\frac{a_1}{a_2} = \frac{b_1}{b_2}$$

$$\Rightarrow \frac{3}{5} = \frac{-a-1}{1-2a}$$

$$\Rightarrow 3 - 6a = -5a - 5$$

$$\Rightarrow -6a + 5a = -5 - 3$$

$$\Rightarrow -a = -8$$

$$\frac{3}{5} \frac{a_1}{a_2} = \frac{c_1}{c_2}$$

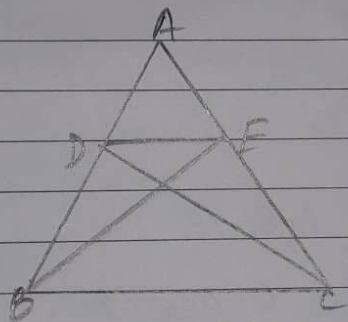
$$\Rightarrow \frac{3}{5} = \frac{2b-1}{3b}$$

$$\Rightarrow 9b = 10b - 5$$

$$\Rightarrow b = 5$$

$\therefore$ For $a = 8$ and $b = 5$ the ^{given} pair of equations have infinite number of solutions.

Ans no 16:- Solⁿ



Solⁿ:- In the given figure $\triangle ABE \cong \triangle ACD$

To Prove = $\triangle ADE \sim \triangle ABC$

Proof:-

$$\triangle ABE \cong \triangle ACD \text{ [given]}$$

$$AE = AD \text{ [by CPCT]}$$

$$AB = AC \text{ [by CPCT]} \rightarrow \textcircled{1}$$

$$AE = AD \text{ [by CPCT]}$$

$$AD = AE \rightarrow \textcircled{2}$$

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(11)
(1)

$$\frac{AD}{AB} = \frac{AE}{AC} \rightarrow (3)$$

$$\text{In } \triangle ADE = \triangle ABC$$

$$\angle A = \angle A \text{ [common angle]}$$

$$\frac{AD}{AB} = \frac{AE}{AC} \text{ [from (3)]}$$

$$\therefore \triangle ADE \sim \triangle ABC \text{ [by SAS]}$$

Hence Proved

Ans no 17:-

LHS

$$\frac{\tan^2 \theta}{\tan^2 \theta - 1} + \frac{\operatorname{cosec}^2 \theta}{\sec^2 \theta - \operatorname{cosec}^2 \theta}$$

$$= \frac{\left(\frac{\sin \theta}{\cos \theta}\right)^2}{\left(\frac{\sin \theta}{\cos \theta}\right)^2 - 1} + \frac{\left(\frac{1}{\sin \theta}\right)^2}{\left(\frac{1}{\cos \theta}\right)^2 - \left(\frac{1}{\sin \theta}\right)^2}$$

$$= \frac{\frac{\sin^2 \theta}{\cos^2 \theta}}{\frac{\sin^2 \theta}{\cos^2 \theta} - 1} + \frac{\frac{1}{\sin^2 \theta}}{\frac{1}{\cos^2 \theta} - \frac{1}{\sin^2 \theta}}$$

$$= \frac{\frac{\sin^2 \theta}{\cos^2 \theta}}{\frac{\sin^2 \theta - \cos^2 \theta}{\cos^2 \theta}} + \frac{\frac{1}{\sin^2 \theta}}{\frac{\sin^2 \theta - \cos^2 \theta}{\cos^2 \theta \cdot \sin^2 \theta}}$$

$$= \frac{\sin^2 \theta}{\sin^2 \theta - \cos^2 \theta} + \frac{\cos^2 \theta}{\sin^2 \theta - \cos^2 \theta}$$

$$= \frac{\sin^2 \theta + \cos^2 \theta}{(\sin^2 \theta - \cos^2 \theta)}$$

$$= \frac{1}{\sin^2 \theta - \cos^2 \theta}$$

= RHS Hence Proved

Point divides the line segment
Ratio $m:n$
Ans no 18:-

Let $C(1,1)$ divides the line segment AB joining points $A(-2,7)$ and $B(x_2, y_2)$ in the ratio $3:2$,

$$C = \left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n} \right)$$

$$\Rightarrow (1,1) = \left(\frac{3x_2 + 2(-2)}{3+2}, \frac{3y_2 + (2 \times 7)}{3+2} \right)$$

$$\Rightarrow (1,1) = \left(\frac{3x_2 - 4}{5}, \frac{3y_2 + 14}{5} \right)$$

$$\Rightarrow \frac{3x_2 - 4}{5} = 1, \quad \text{and} \quad \frac{3y_2 + 14}{5} = 1$$

$$\Rightarrow 5 = 3x_2 - 4, \quad 5 = 3y_2 + 14$$

$$\Rightarrow 3x_2 = 9, \quad 3y_2 = -9$$

$$\Rightarrow x_2 = 3, \quad y_2 = -3$$

$\therefore$ The point $B(x_2, y_2) = 3, -3$

Point divides the line segment
Ratio $m:n$
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$$C = \left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n} \right)$$

$$\Rightarrow (1,1) = \left(\frac{3x_2 + 2(-2)}{3+2}, \frac{3y_2 + (2 \times 7)}{3+2} \right)$$

$$\Rightarrow (1,1) = \left(\frac{3x_2 - 4}{5}, \frac{3y_2 + 14}{5} \right)$$

$$\Rightarrow \frac{3x_2 - 4}{5} = 1, \quad \text{and} \quad \frac{3y_2 + 14}{5} = 1$$

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$\therefore$ The point $B(x_2, y_2) = 3, -3$

(14)

$$\triangle ABC \sim \triangle DEC$$

Ans no 19: Solve

Let the numerator be x and the denominator be y

$$\therefore \text{The fraction} = \frac{x}{y}$$

Acc. to 1st condition

$$\frac{2x}{y+1} = 1$$

$$\Rightarrow 2x = y + 1$$

$$\Rightarrow 2x - y = 1 \rightarrow \textcircled{I}$$

Acc. to 2nd condition

$$\frac{x+4}{2y} = \frac{1}{2}$$

$$\Rightarrow 2x + 8 = 2y$$

$$\Rightarrow 2x - 2y = -8$$

$$\Rightarrow x - y = -4 \rightarrow \textcircled{II}$$

$$\textcircled{I} + \textcircled{II} \quad \textcircled{I} - \textcircled{II}$$

$$2x - y - (x - y) = 1 - (-4)$$

$$\Rightarrow x = 1 + 4$$

$$x = 5$$

Putting $x = 5$ in $\textcircled{II}$

$$5 - y = -4$$

$$\Rightarrow -y = -4 - 5$$

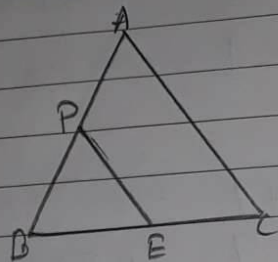
$$y = 9$$

∴ The fraction = $\frac{5}{9}$

Quesno 20: Soln

Basic Proportionality theorem

- If a line is drawn parallel to one side of a triangle intersecting the other two sides in distinct points, then it divides the two sides in the same ratio.



Soln: A line PE is drawn parallel to AC of $\triangle ABC$ intersecting AB and BC at P & E respectively.

To Prove :- $\frac{AP}{PB} = \frac{AE}{EC} = \frac{CE}{BE}$

Proof :-

∵ $PE \parallel AC$ and BC transversal
 $\angle PEB = \angle ACB$ [corresponding angles]
 $\rightarrow \textcircled{1}$

In $\triangle PEB$ and $\triangle ACB$

In $\triangle ACB$ and $\triangle PEB$

$\angle ABC = \angle PBE$ [common angle]

$\angle ACB = \angle PEB$ [From $\textcircled{1}$]

∴ $\triangle ACB \sim \triangle PEB$ [by AA postulate]

$$\frac{AB}{PB} = \frac{BC}{BE}$$